

Measuring Drift in Long Reading Sessions

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Abstract

We describe a small observational study of attention drift while reading technical papers on screen. Readers were asked to follow a single argument across several pages while we recorded where they paused, where they went back, and where they gave up. The results suggest that interruptions cluster around page turns and figure references rather than around difficult passages.

1 Introduction

Reading a research paper is rarely a linear activity. A reader moves back to an earlier definition, jumps ahead to a figure, and returns to the paragraph that cited it, often several times per page. Each of these moves carries a small cost, and the costs add up over a long session in ways that are easy to overlook when a tool is designed around single page views.¹

Earlier work on reading interfaces has mostly measured speed and recall on short passages. Those measures are useful, but they do not capture what happens in the fortieth minute of a dense proof, when the reader has already flipped between the same two pages a dozen times and the thread of the argument has started to fray.

1.1 Scope of this study

We restrict attention to papers typeset in a single column with numbered sections, because that layout makes page turns the dominant kind of interruption. Two-column layouts introduce a second kind of jump, from the bottom of one column to the top of the next, which we leave for a separate report.

¹We use the word session for one uninterrupted sitting with a single document.

2 Method

Each participant read one paper of about twenty pages while an observer logged every backward jump. We summarise a session by its drift score, which weights each jump by the distance travelled and discounts jumps that end on a figure or table:

$$D = \frac{1}{T} \sum_{k=1}^n w_k |p_k - p_{k-1}|. \quad (1)$$

Here the sum runs over all jumps in a session of length T minutes, and the weight is smaller for jumps that land on a float.

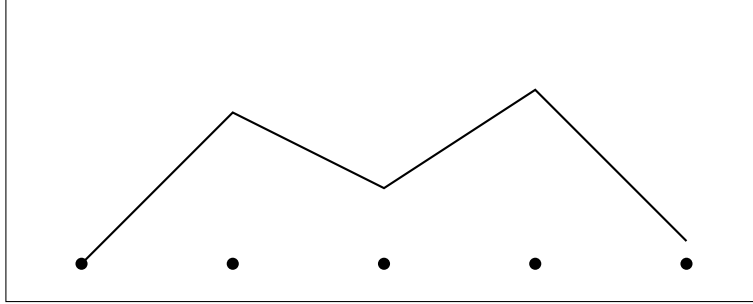


Figure 1: Drift score over time for one representative session. The score rises sharply after each page turn and decays slowly afterwards.

2.1 Participants and materials

Twelve graduate students took part. Each chose a paper from their own field that they had not read before, so that the material was neither trivial nor entirely unfamiliar. Sessions were held in a quiet office with the same monitor and the same reader software for everyone.

3 Discussion

The most striking finding is how little the difficulty of a passage predicted drift. Readers stayed with a hard derivation for a long time and moved on smoothly once it was done. The flow broke instead at points that look trivial on paper: a figure placed two pages after its first mention, a definition that was only given in a footnote, or an efficient but terse notation introduced without warning. We think these are exactly the places where a reader benefits from a quick preview rather than a full jump to another part of the document, because the representational cost of leaving the current page is paid whether the detour takes one second or one minute. The effect was sufficiently consistent across participants that we regard it as the main result.

3.1 Limitations

The sample is small and self-selected, and a single observer logged all of the sessions, so some jumps were probably missed. We also cannot separate the influence of the paper from the influence of the field, since every participant chose a paper from their own discipline. A controlled follow-up with a fixed reading list would resolve this, at the price of less natural reading behaviour and a somewhat artificial task.

3.2 Future work

We plan to repeat the study with an instrumented reader that records jumps automatically and offers inline previews of figures, equations and cited works. Comparing drift with and without previews would tell us whether the cost we measured is intrinsic to the material or an artefact of the interface, and whether the benefit persists once the novelty wears off. We also intend to examine whether the effect changes for readers who are unfamiliar with the notational conventions of the field, since for them even a short detour may involve a chain of further look-ups.

4 Conclusion

Drift during long reading sessions is driven more by layout than by difficulty. Tools that shorten the round trip between a reference and its target should help most where the layout forces the reader to leave the page.